

Hand-Controlled Drone — Bill of Materials

Tools

Item	Spec	Qty needed	Buy	Est. Price	Link
Soldering iron kit (all-in-one)	60W adjustable temp, includes digital multimeter, wire stripper/cutter, solder wire, desoldering pump, tweezers, stand	1	1	~\$27	Soldering Iron Kit w/ Digital Multimeter, Wire Stripper, Solder

This single kit covers the soldering iron, solder, wire stripper, and multimeter all at once — no need to buy those separately.

Flight Electronics (center stack)

Item	Spec	Qty needed	Buy	Est. Price	Link
ESP32 Dev Board	ESP-WROO M-32	2 (1 drone, 1 hand controller)	2-pack	~\$12	HiLetgo 2pcs ESP32
IMU	MPU6050 (GY-521)	2 (1 drone, 1 glove)	2-pack	~\$8	Ximimark 2pcs MPU-6050
Radio	nRF24L01+P A+LNA (SMA antenna)	2 (1 pair)	2-pack	~\$15	HiLetgo 2pcs nRF24L01+P A+LNA

Propulsion

Item	Spec	Qty needed	Buy	Est. Price	Link
Brushless motors	2205 2300KV, 2×CW + 2×CCW	4	1 set of 4	~\$32	Readytosky RS2205 2300KV (4pcs)
4-in-1 ESC	BL_S 30x30	1 board (drives all 4 motors)	1	~\$32	AERO SELFIE 45A 4in1 ESC
Propellers	5045 tri-blade, 2×CW + 2×CCW	4 (+spares)	2 packs (buy spares — you will crash)	~\$16 for 2 packs	Gemfan 5045 propellers (2 pair)

Power

Item	Spec	Qty needed	Buy	Est. Price	Link
LiPo battery	4S 1500mAh, 100C, XT60	2 (fly one, charge the other)	2-pack	~\$35	CNHL 1500mAh 4S 100C (2 pack)
Balance charger	1S–6S, 80W/6A	1	1	~\$25	80W 6A Balance Charger/Discharger
XT60 connectors	male/female pairs, for wiring battery leads/PDB	~5 pairs	20-pair pack (spares)	~\$10	XT60 Connectors, 20 pairs w/ heat shrink
Low-voltage alarm	plug-in LiPo cell checker/buzzer (no coding needed)	1	4-pack	~\$9	Readytosky LiPo Battery Checker/Alarm (4pcs)

Item	Spec	Qty needed	Buy	Est. Price	Link
PDB	3S-10S Input, 5V/12V Output	1	1	~\$24	AERO SELFIE 4IN1 PDB
Buck Converter	3v to 5v	1	2 pack	~\$5	Buck Converter Mini 5V 3V

Build / Assembly Hardware

Item	Spec	Qty needed	Buy	Est. Price	Link
Perfboard	for soldering ESP32/MPU 6050/nRF24L 01 in place of a PCB	a few boards	32-pc assorted kit	~\$9	ELEGOO 32pc Prototype Board Kit
Nylon standoffs	M3, assorted lengths, for stacking plates	as needed	320-pc kit	~\$10	Litorange 320pc M3 Nylon Standoff Kit
Silicone hookup wire	22AWG, multi-color, for point-to-point wiring	1 kit	1	~\$13	TUOFENG 22AWG Silicone Wire (6 colors)
Vibration-da mping foam	isolates flight electronics from motor vibration	1 pack	12-pc	~\$8	CSM Vibration Isolation Foam Squares (12pc)
Heat shrink tubing	assorted sizes, for wire joints	1 kit	—	~\$8	<i>(included w/ some connector kits above, or buy any</i>

Item	Spec	Qty needed	Buy	Est. Price	Link
					<i>assorted heat-shrink pack)</i>

Hand Controller (glove/handheld unit)

Reuses the second ESP32 + MPU6050 + nRF24L01 listed above. Mount them on:

- A **snug fingerless glove or elastic wrist strap** (any sporting-goods or hardware store option works — no special electronics-specific product needed; a wristband is actually easier to mount rigidly than a glove under time pressure)
- Small piece of leftover perfboard + double-sided foam tape to fix the IMU flat against the back of the hand/wrist so its orientation stays consistent

Frame

Custom two-tier stacked frame with arms — 3D printed (handled separately).